EIE4432

Web Systems and Technologies

Project Report

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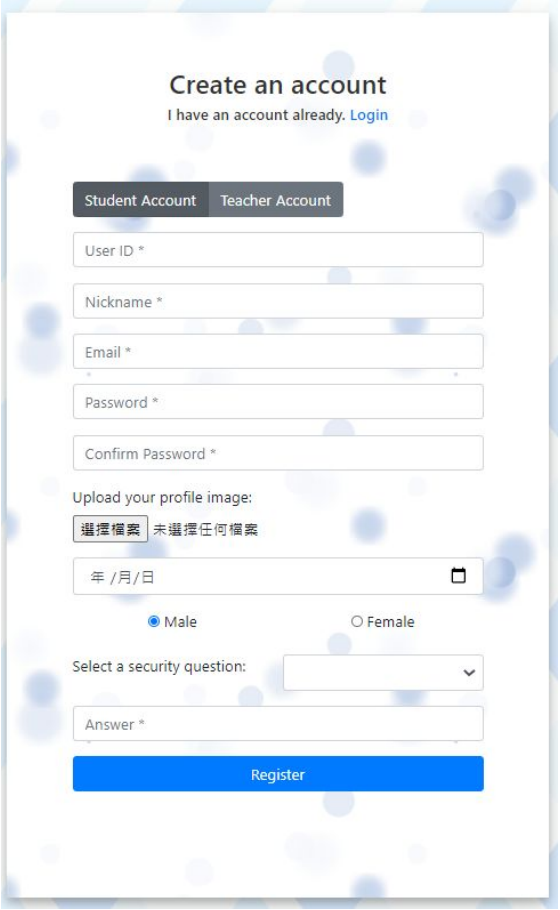
Introduction

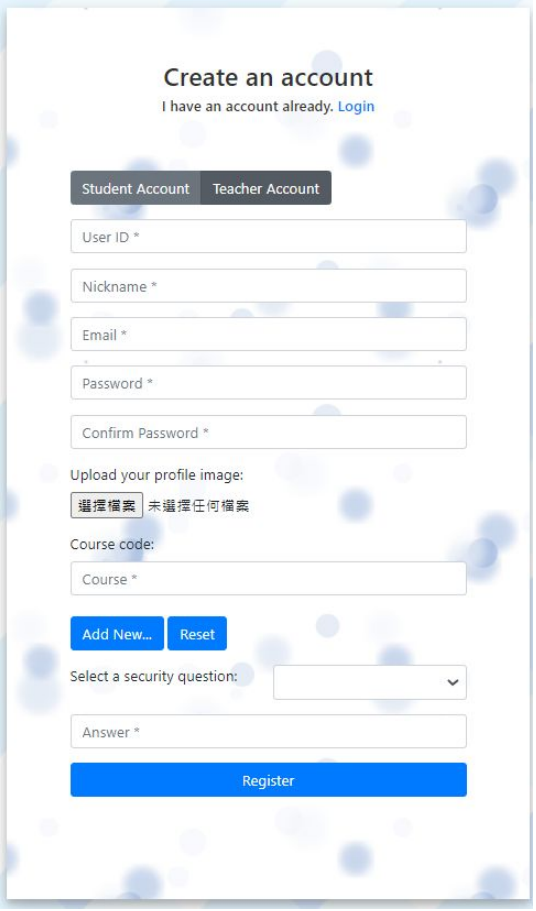
This milestone project is an online examination system site where teachers can release an exam and evaluate the student's results, and students can take an exam, and the system administrator can manage the information of the teacher and students. In this project, we mainly used Apache, PHP and MySQL as web application infrastructure to construct the site, with the assist of javascript and HTML. Powerful APIs such as JQuery and Bootstrap are also used in the frontend to provide better user interface and user experience. We used Github to do the version control and used Visual Studio Code live share to achieve real-time collaboration. In this report, all the functions and the database design will be presented.

Functions Explanations

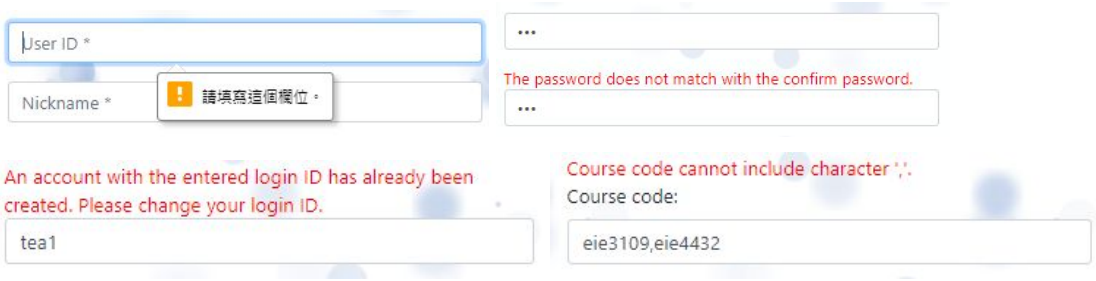
User Registration

User registration for the two roles are done within the same page, the user can switch in between by pressing the button on top.





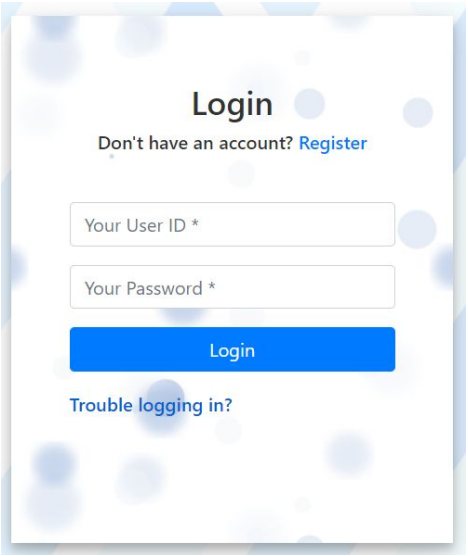
Pressing “Add New...” can add new text fields without affecting the others for inputting new course codes, while “Reset” will clear all text field and only leave one course code field. This is achieved by modifying the innerHTML through javascript. The registration process is strictly supervised with validation checks, shown as below.



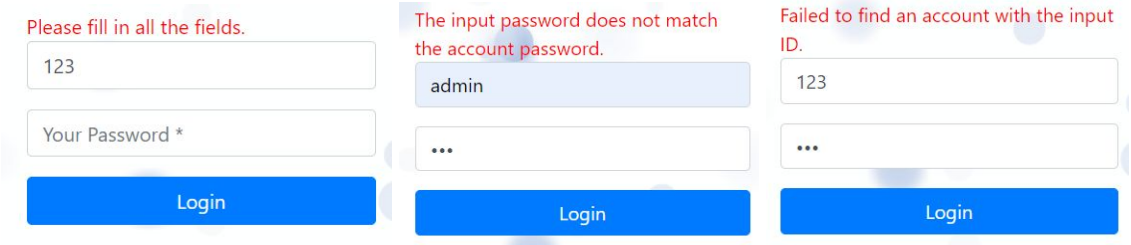
Every visible data field is checked through HTML to confirm they are all filled up before registration. And the inputted user ID will be checked with the table “users” to see if there is already an account with the same ID through AJAX . The course code cannot contain the character “,” which is crucial to the interpretation of course in the database table.

The upload button for the profile image restricts user to only choose the image file types. However if the user did upload a file that is not an image, the system would indicate this as a failed registration and tell user to redo registration with a proper file type. The php file responsible for registration will check the file type before uploading. Also the uploaded image is renamed with the user ID and put into the same file, for avoiding duplicate image names and easier management.

User Login



User login finds an account with the matching input ID and input password using AJAX.



After validation, the web system will record the nickname, userID and user type(student, teacher or admin) in the cookies and redirect the user to the corresponding web page. Every time the user enters a new page, the cookie will be recorded again for 6 hours. When user logout, the cookie is deleted and user is redirected back to the login page. The following is the cookie part of the code used when login, logout and renew cookie.

```
//save data, record cookie for 6hours
setcookie("type", $type, time() + 21600, '/');
setcookie("userID", $ID, time() + 21600, '/');
setcookie("nickName", $nick, time() + 21600, '/');
//login success - Request.responseText to checklogin.js
echo $type;
```

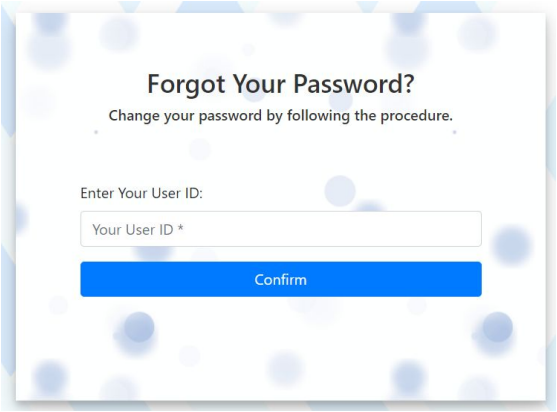
```
if (isset($_COOKIE["type"])){
 setcookie("type", $_COOKIE["type"], time() - 21600, "/");
}
if (isset($_COOKIE["userID"])){
 setcookie("userID", $_COOKIE["userID"], time() - 21600, "/");
}
if (isset($_COOKIE["nickName"])){
```

```
 setcookie("nickName", $_COOKIE["nickName"], time() - 21600, "/");
}
$alert_message = "You have logged out!";
$link = "../login.html";
```

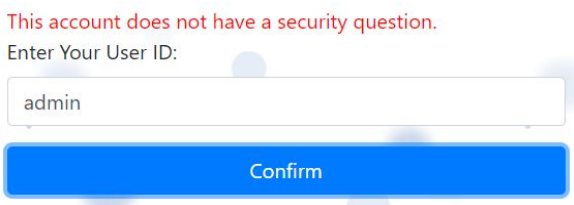
```
if (isset($_COOKIE["type"])){ //check if cookie still exist
  //renew the cookie
 setcookie("type", $_COOKIE["type"], time() + 21600, "/");
 setcookie("userID", $_COOKIE["userID"], time() + 21600, "/");
 setcookie("nickName", $_COOKIE["nickName"], time() + 21600, "/");
```

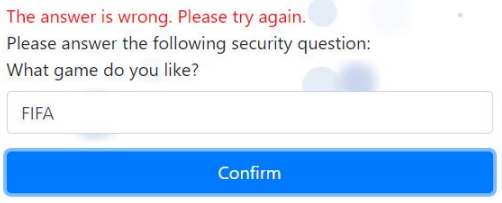
For the logout button, user can always press the button at top right of the page to log out, unless in some sections(e.g. during the student answering a paper).

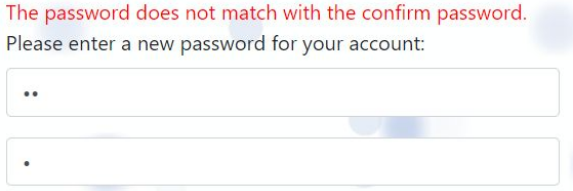
We also implemented a forget password section for user by letting them answer their security question. After completing the section, the user can reset the password of their account.



This section is also monitored by validation. As for special accounts like admin, they do not have a security question set, so resetting their password is impossible.

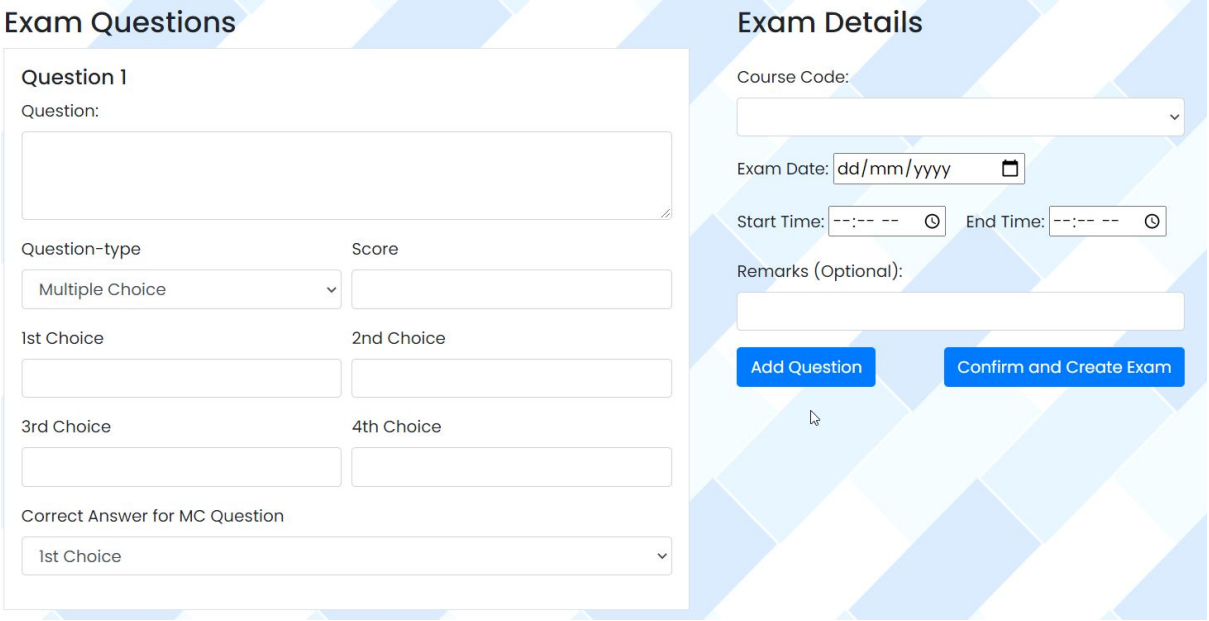


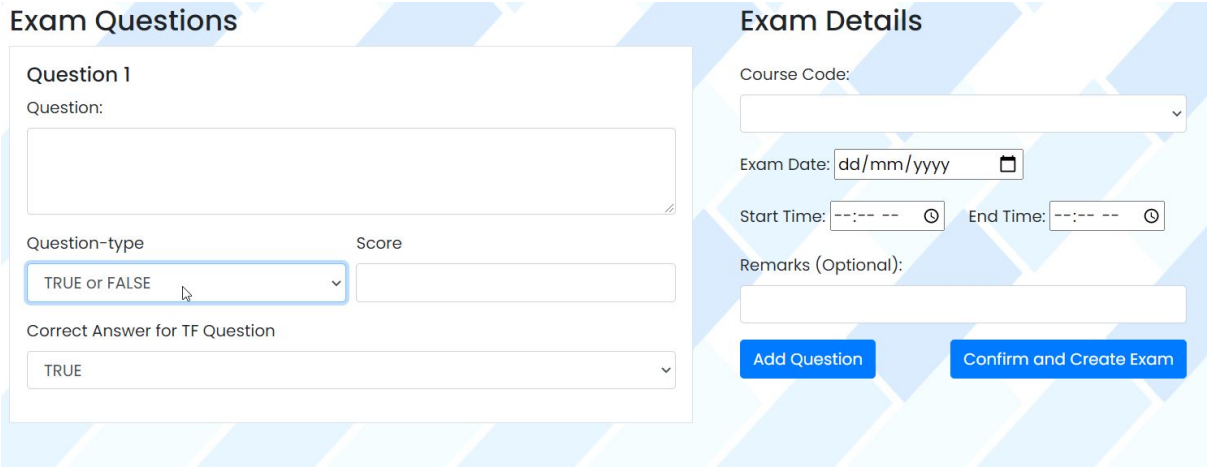


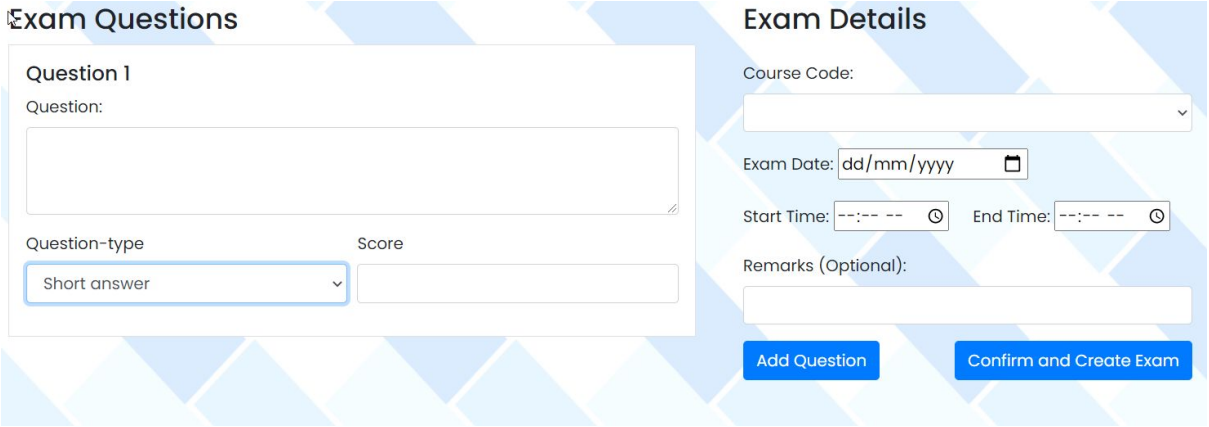


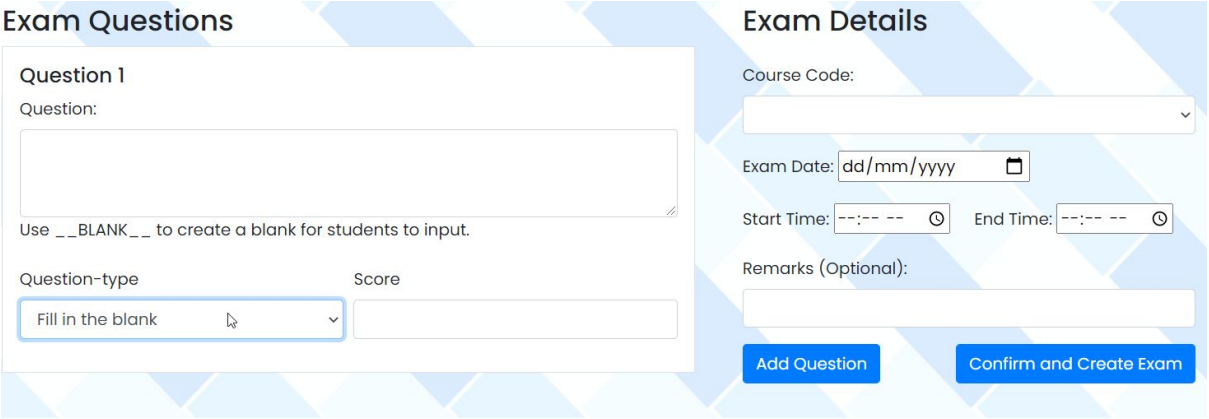
Teacher Releases an Exam

Teacher account can create an exam by specifying the time slot and the questions. The teacher can add as many questions as he/she likes. We support 4 types of questions, including Multiple Choices, True False Questions, Fill in the blank and Short Answer. For each question, after the user specifies the question type, the matching inputs will be shown. This is achieved by hiding and showing some input elements according to selected Question-Type.

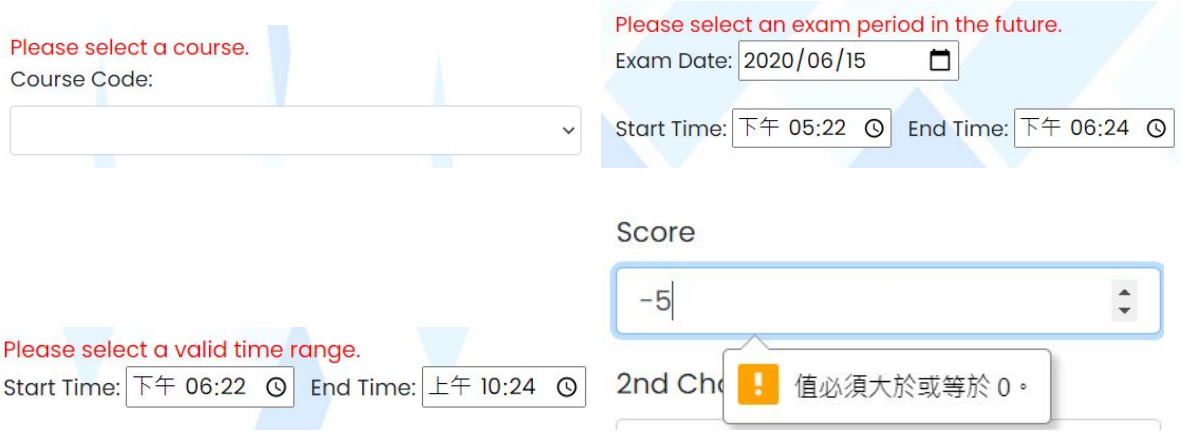








After setting up the questions, the teacher can press the confirm and create exam button to release the exam to students. The questions will be set into the `exam_questions` table in the database while the general information will be set inside `exams` table. Validation is applied to ensure the teacher created a proper exam.



The exam date and time should be set at the time of creating exam or after it, meanwhile the start time should not be set as the same or before the end time within that day. Also, necessary data fields in the question paper, e.g.score, question and correct answer, should be filled in. The score should be an positive integer.

Teacher Evaluates and Views Students' Submitted Answers

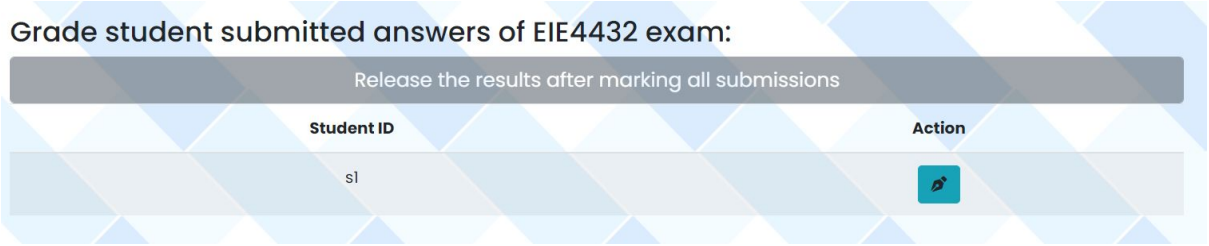
Since we have chosen to add other question types (i.e. fill in the blank, short answer), we also implemented the page for teachers to check and grade the submitted answer. In that page teacher can also view the exam result of each student (questions, submitted answers, correct answers, total score, score per question, submitted time). Teacher must grade all the students' submitted answers before releasing the results. If all the questions are MCQ or TFQ, the teacher need not to manually grade the students' submitted answers.

Teacher can view the ungraded exams created by them through a table. A message will be shown instead if no exams are available for grading.

Created Exams

Course Code	Exam Time	Date	Remarks	Action
EIE3320	07:00-08:00	2020-12-13	-	
EIE3320	09:00-12:00	2020-12-13	part 2	
EIE3320	20:30-20:45	2021-05-30	Next Year	
EIE4432	17:00-18:30	2020-10-11	Open-Book	
EIE4432	18:15-19:30	2020-12-30	Re-examination	

After selecting the exam waiting to be graded, teacher will enter a page in which the ungraded student attempts on that exam will be listed.

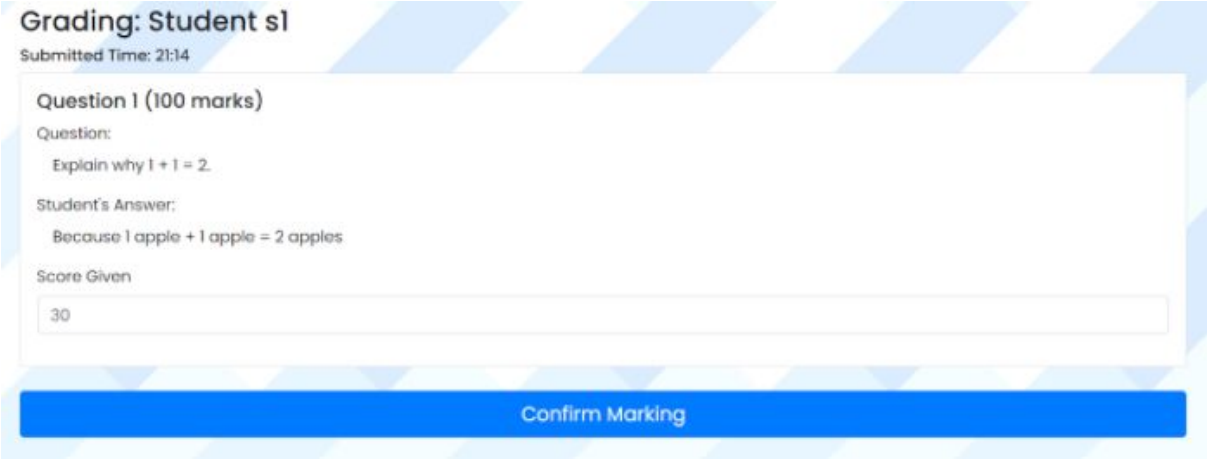


Grade student submitted answers of EIE4432 exam:

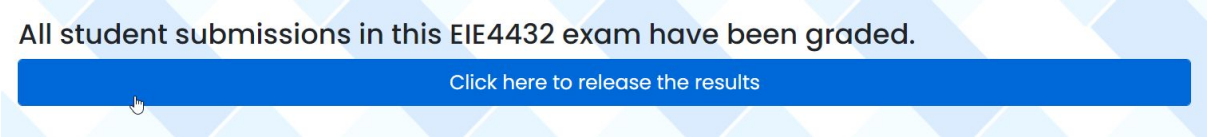
Release the results after marking all submissions

Student ID	Action
s1	

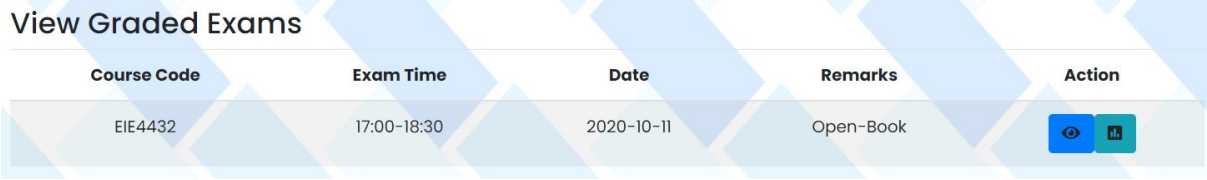
The teacher can grade the answer sheet by clicking the button on the right.



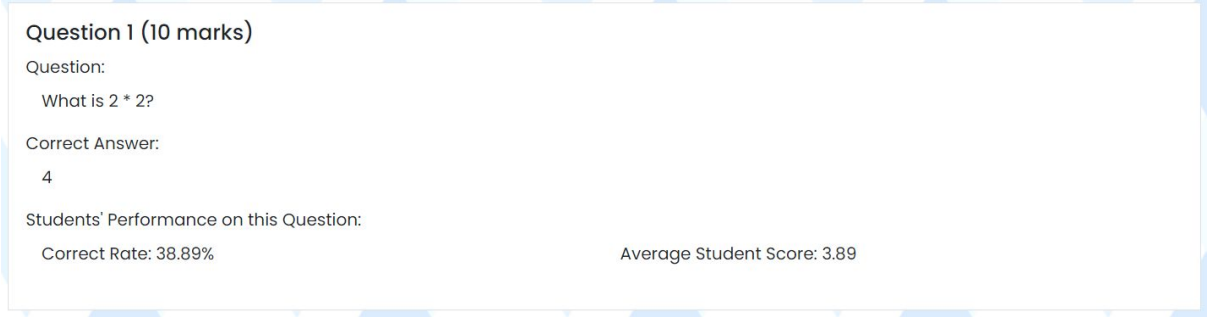
Teacher can then give score to the student answer by entering the text field. Note that teacher can only enter score between zero and maximum score of the question. After grading all submissions, the exam table will show no student attempts. Teacher can release the results to the students, the statistics of that exam will then be calculated.

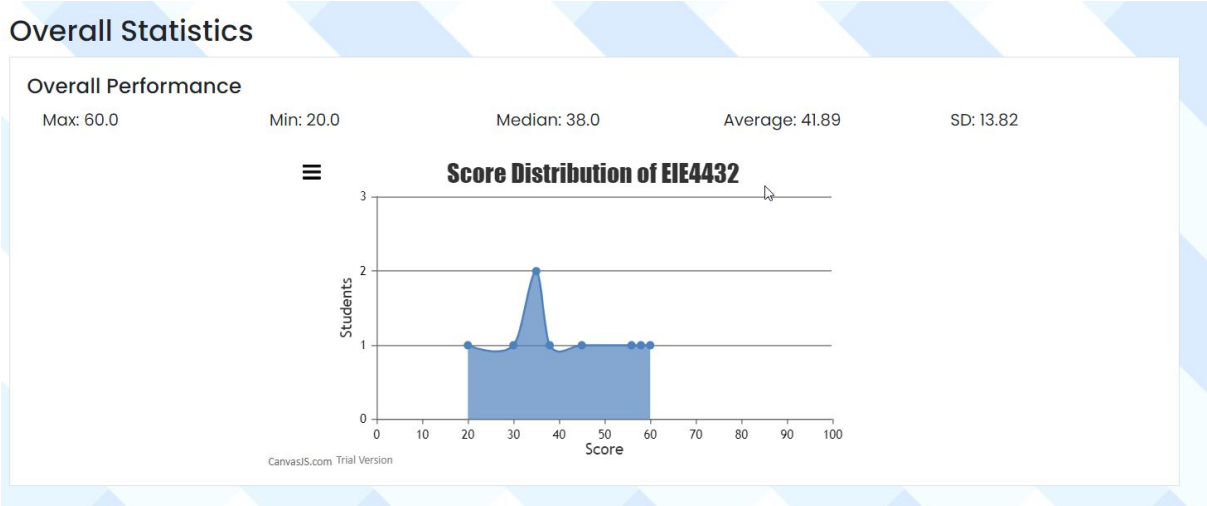


The overall statistic and individual submissions of an exam can be viewed once published.



Clicking the statistic button will allow teacher to see the statistics of the exam score (max, min, median, average, standard deviation), and attempt statistics (percentage of correct answer and average score of each question).





The file math-functions.php and teacher-get-statistics.php are responsible for handling overall statistics of the exam. The score distribution can be exported as an image. Here showcasing some of the functions to compute the statistics:

```
function getMedian($arr) {
        if(!is_array($arr)){
            throw new Exception('$arr must be an array!');
        }
        if(empty($arr)){
            return false;
        }
        $num = count($arr);
        $middleVal = floor(($num - 1) / 2);
        if($num % 2) {
            return $arr[$middleVal];
        }
        else {
            $lowMid = $arr[$middleVal];
            $highMid = $arr[$middleVal + 1];
            return (($lowMid + $highMid) / 2);
        }
    }
```

```
    function getAverage($arr) {
        if(!is_array($arr)){
            throw new Exception('$arr must be an array!');
        }
        if(empty($arr)){
            return false;
        }
        else
        {
            $avg = array_sum($arr) / count($arr);

            return $avg;
        }
    }
```

Individual submissions of the participants can also be viewed by the teacher. Upon choosing which student to view, the system will extract the data of the answers from that student in this exam.

View graded students of EIE4432 exam:

Student ID	Action
gary315	
ghergher	
hahahajustin	
harvardU	
peener	
s2	
s3	
sarsarsar	
tete	

Viewing: Student gary315

Submitted Time: 18:30

Question 1 (10 marks)

Question:

What is 2 * 2?

Student's Answer:

4

Correct Answer:

4

Score Given

10

Question 6 (10 marks)

Question:

Everyday I go to school by _____.

Student's Answer:

NOT ANSWERED

Score Given

0

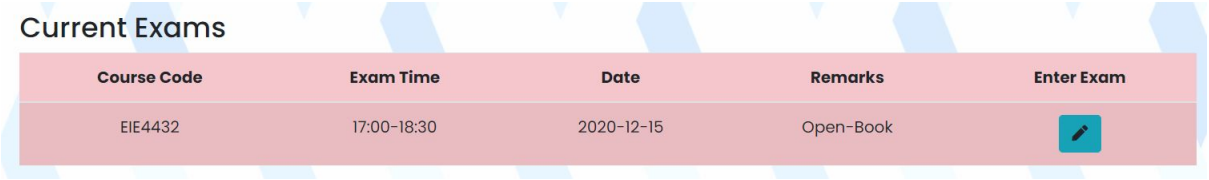
Student's Total Score

45

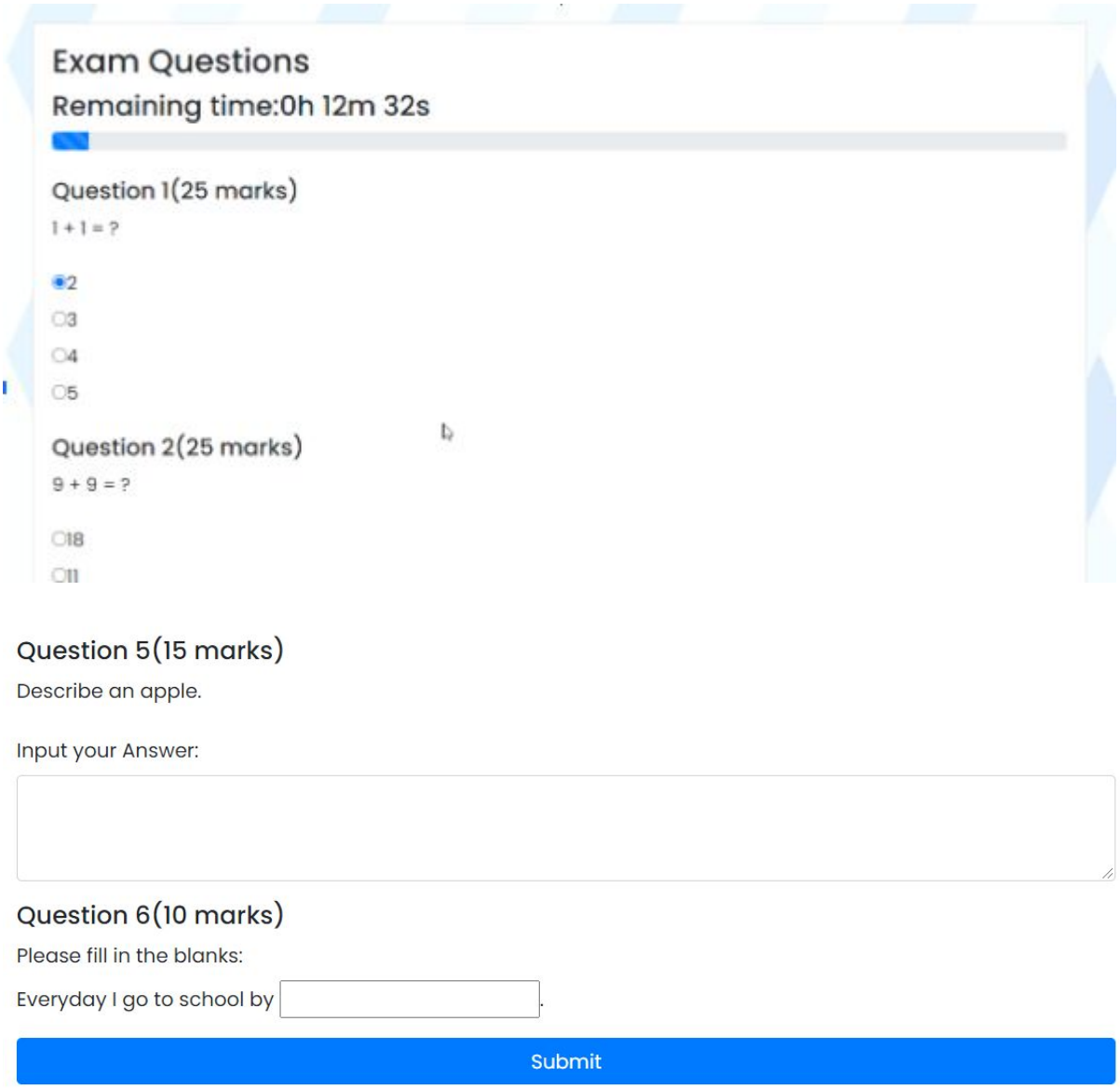
Full Score

100

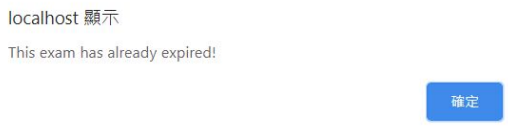
Student takes an exam

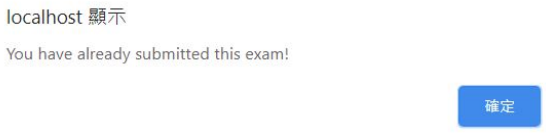


Once reaching the exam time, students would go to the Take Exam page and choose the exam to start answering questions. If reaching the expiration time, the student's answers will be saved and submitted automatically.

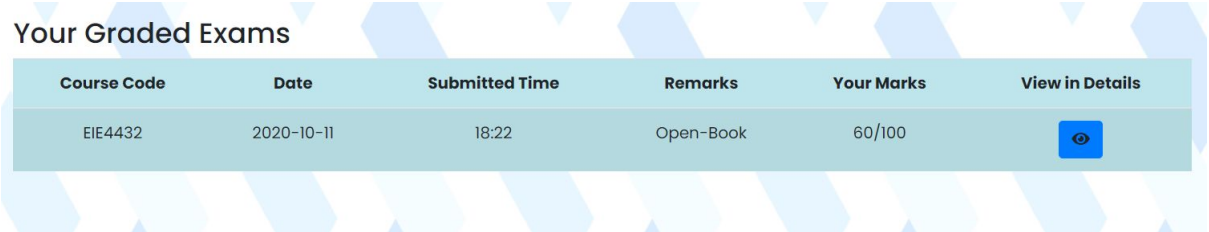


Once the student submitted his attempt or the expire time reaches, the student cannot enter the same examination again.





After the teacher releases the grades of the exam, students can see the marks and details in the page View Result . The details include questions, submitted answers, correct answers, total score, score per question and submitted time. Basically what is shown in teacher's view of individual submission will be shown.



Your Graded Exams

Course Code	Date	Submitted Time	Remarks	Your Marks	View in Details
EIE4432	2020-10-11	18:22	Open-Book	60/100	

System Management

If an admin account is logged in, the admin can see a page where all the user's information and general information of all examinations is located. The records are stored in a table. In each teacher/student account record there are two action buttons next to it. Admin can either modify or delete the corresponding record.

Add a new teacher or student account

Admin Accounts

ID	Password	Nickname	Email	Profile Image
admin	admin4432	admin	systemadmin@gmail.com	

Teacher Accounts

ID	Password	Nickname	Email	Profile Image	Course		
tea1	123	Dr. Kee	mrkee@poly.com		EIE4432,EIE3320		
www	12311	Dr. Lee	drlee@poly.com		EIE3109		

Student Accounts

ID	Password	Nickname	Email	Profile Image	Gender	Birthday		
gary315	313325	Gary	gary@wow.com		M	2010-07-31		
ghergher	456654	Gherman	gherr@yooo.com		M	2012-01-27		
hahahajustin	173537	Justin	justin@hahaha.com		M	2008-06-30		
harvardU	12346	Havard	havard@yee.com		M	2007-04-30		

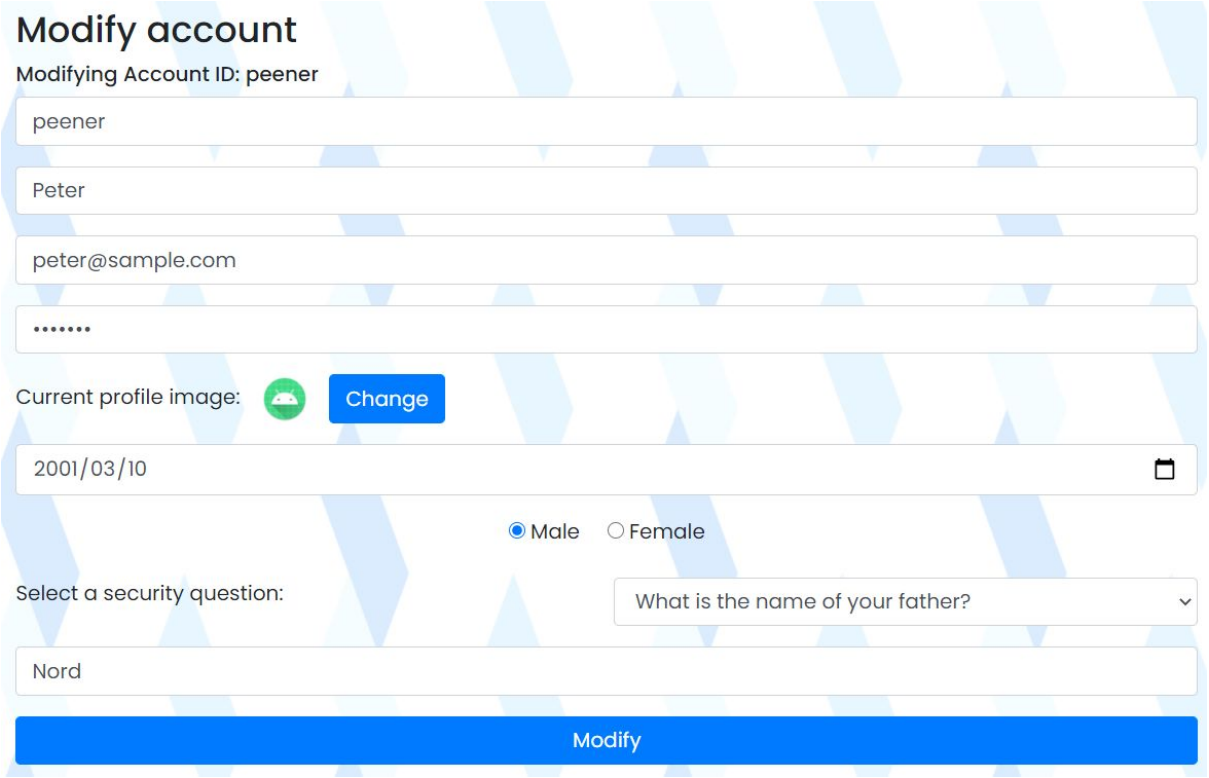
Examination Info

Course Code	Exam Number	Exam Date	Start Time	Expire Time	Creator	Remarks	Status
EIE3109	1	2020-12-14	09:00	12:00	mrlee	By Dr.Lee!	Not Published
EIE3320	1	2020-12-13	07:00	08:00	tea1	-	Published
EIE3320	2	2020-12-13	09:00	12:00	tea1	part 2	Not Published
EIE3320	3	2021-05-30	20:30	20:45	tea1	Next Year	Not Published
EIE4432	1	2020-12-15	17:00	18:30	tea1	Open-Book	Published
EIE4432	2	2020-12-30	18:15	19:30	tea1	Re-examination	Not Published

The admin can add/modify/remove a teacher/student account. To achieve add, add-account.php is used (sharing the same core with registration page). A page with similar design of registration page will pop up for admin.

For account deletion, it is handled by admin-delete-account.php. The account record will be deleted in the `users` table and the page will be refreshed.

Account modification is achieved by admin-modify-account.php, when an account is selected to modify, admin will then go to a page to modify its information, with the account's original data inputted in the text fields automatically.



Any data input cannot be empty, while the user ID will be checked with the `users` table to see if the ID already existed(only when it is changed). Changing the profile image will overwrite the original image file.

Additional Functions

Responsive Design

A responsive web design will automatically adjust for different screen sizes and viewports.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

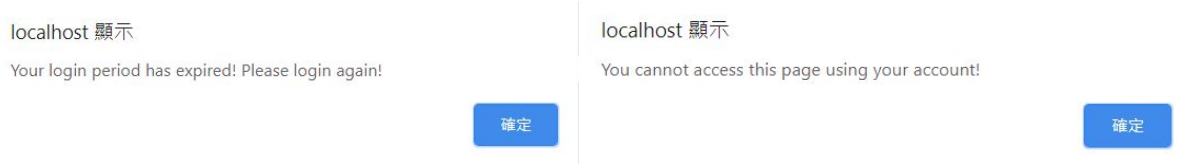
Time bar

The time bar shows the overall time passed for the exam period since the start time of the exam. The progress and the timer is updated in real time for student to view.

Auto login, redirect and restrictions

We want to fully utilized the use of cookies, so we also use it to redirect users. When a logined user goes to the login page, he will be redirected back to the dashboard of the logged account without the need of login again.

Everytime he goes to another page, the cookie will be checked to see if its login status is expired or not. If it is expired, it will timeout the user and force him to logout. If it is not, the cookie will be recorded again for 6 hours. The system will also check if the account type(student, teacher or admin) is authorized to access the target page or not. If not, the user will be redirected back to its dashboard page.



Back to Top button

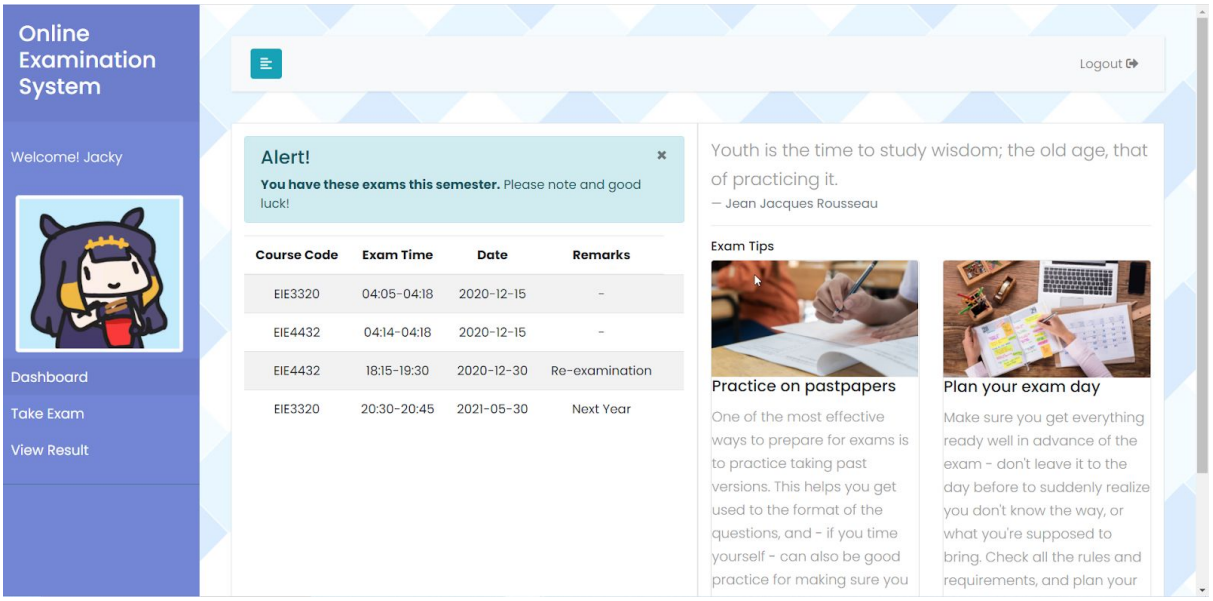
For pages that might result in long scrolling, we added a button to let the user quickly move back to the top. The button is located in the right bottom of the page (fixed location), and it appears when long scrolling action is detected. The implementation is in back-to-top-btn.php.

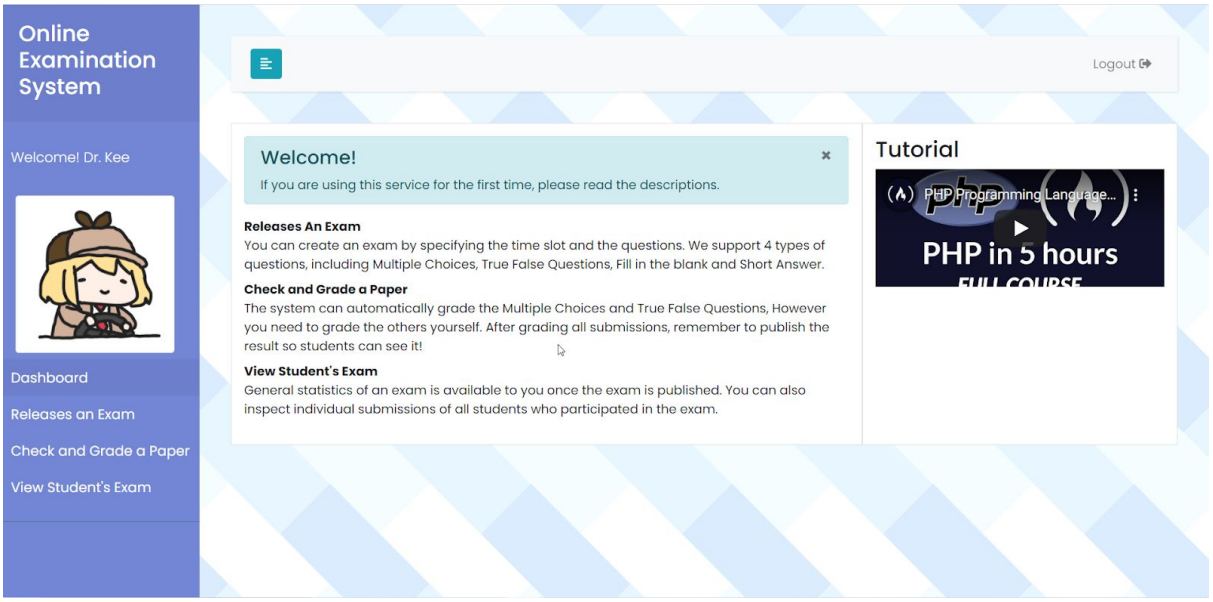
```
<a id="back-to-top" href="#" class="btn btn-light btn-lg back-to-top" role="button"><i class="fas
fa-chevron-up"></i></a>
<style>
   .back-to-top {
   position: fixed;
   bottom: 25px;
   right: 25px;
   display: none;
   }
</style>
<script>
   //Back to Top Button using JQuery
   $(document).ready(function(){
        $(window).scroll(function () {
                        if ($(this).scrollTop() > 50) {
                                $('#back-to-top').fadeIn();
                        } else {
                                $('#back-to-top').fadeOut();
                        }
                });
                // scroll body to 0px on click
                $('#back-to-top').click(function () {
                        $('body,html').animate({
                                scrollTop: 0
                        }, 400);
                        return false;
                });
   });
</script>
```

Dashboard & Sidebars features

After the user logged in, the user can interact with the sidebar to navigate to other pages. The sidebar can be hidden by pressing the top left button. The side bar also contains a small profile of the user.

In order to make the Transitions smoother, a Dashboard is added in both student and teacher's account. The student's dashboard will show the information of upcoming exams, random motivation quotes and some exam tips. While the teacher's dashboard will show how to use the online examination system and there is video support on how to use the examination system (the PHP tutorial as placeholder).


Online Examination System
Welcome! Jacky
Dashboard
Take Exam
View Result
Logout
Alert!
You have these exams this semester. Please note and good luck!
Course Code | Exam Time | Date | Remarks
EIE3320 | 04:05-04:18 | 2020-12-15 | -
EIE4432 | 04:14-04:18 | 2020-12-15 | -
EIE4432 | 18:15-19:30 | 2020-12-30 | Re-examination
EIE3320 | 20:30-20:45 | 2021-05-30 | Next Year
Youth is the time to study wisdom; the old age, that of practicing it.
— Jean Jacques Rousseau
Exam Tips
Practice on pastpapers
One of the most effective ways to prepare for exams is to practice taking past versions. This helps you get used to the format of the questions, and - if you time yourself - can also be good practice for making sure you
Plan your exam day
Make sure you get everything ready well in advance of the exam - don't leave it to the day before to suddenly realize you don't know the way, or what you're supposed to bring. Check all the rules and requirements, and plan your


Online Examination System
Welcome! Dr. Kee
Dashboard
Releases an Exam
Check and Grade a Paper
View Student's Exam
Logout
Welcome!
If you are using this service for the first time, please read the descriptions.
Releases An Exam
You can create an exam by specifying the time slot and the questions. We support 4 types of questions, including Multiple Choices, True False Questions, Fill in the blank and Short Answer.
Check and Grade a Paper
The system can automatically grade the Multiple Choices and True False Questions, However you need to grade the others yourself. After grading all submissions, remember to publish the result so students can see it!
View Student's Exam
General statistics of an exam is available to you once the exam is published. You can also inspect individual submissions of all students who participated in the exam.
Tutorial
PHP Programming Language...
PHP in 5 hours

Database Design

The database of the online examination system consists of four tables: `users`, `exams`, `exam_questions` and `exam_answers`. All SQL operations are done through SQL prepared statements to avoid SQL injection.

users

`users` is a table dedicated to record the user account information of all users. The following is the statement used to construct this table:

```
$createTable1 = "CREATE TABLE users (
  ID varchar(64) NOT NULL,
  userType varchar(64) NOT NULL,
  PW varchar(64),
  nickName varchar(64),
  email varchar(64),
  profileImage varchar(255),
  course varchar(64),
  gender varchar(1),
  birthday date,
  sQType int,
  sQAnswer varchar(64),
  PRIMARY KEY (ID),
  UNIQUE KEY ID (ID)
 )";
```

Data Field	Usage and Description
ID	This is used to uniquely identify all accounts, and it is used to represent the user in all other tables of the database.
userType	This is used to indicate the type of account(student, teacher or admin). This is critical information which the system uses to determine the user's right to access the pages and redirect them to appropriate sections.
PW	This is used to record the password of the account. Password is needed for a user to login the account.
nickName	This is used to record the nickname of the user. The nickname is displayed throughout user activity in the website at the sidebar.
email	This is used to record the email address of the user.
profileImage	This is used to store the name of the image file for the user's profile image. It can be linked to the profile image uploaded to the server file "img/client/". The name of the file is maintained in the format of (userID + file type), e.g."s1.png".

course	This is used to record the course code of the teacher responsible for. This data field may store multiple courses, as the system identifies “,” as a separator of entered courses. For any account with user type other than “teacher”, it is left as NULL.
gender	This is used to record the gender of the student. For any account with user type other than “student”, it is left as NULL.
birthday	This is used to record the birthday of the student. This data field is recorded in a string format of (yyyy-mm-dd), e.g.“2000-12-16”. For any account with user type other than “student”, it is left as NULL.
sQType	This is used to record the type of security question the user chose. Different integers refers to different questions, which is translated by the system when in use.
sQAnswer	This is used to record the user’s answer for the security question.

exams

`exams` is a table for storing the general information of all created exams. The following is the statement used to construct this table:

```
$createTable2 = "CREATE TABLE exams (
        course varchar(64) NOT NULL,
        examNum INT NOT NULL,
        examDate date NOT NULL,
        startTime varchar(64) NOT NULL,
        expireTime varchar(64) NOT NULL,
        creator varchar(64) NOT NULL,
        graded INT(1) DEFAULT 0 NOT NULL,
        remarks varchar(64),
        maxScore double(10,2),
        minScore double(10,2),
        median double(10,2),
        average double(10,2),
        PRIMARY KEY (course, examNum)
   )";
```

Data Field	Usage and Description
course	This is used to identify the course of this exam is designed for. It is used together with “examNum” to uniquely identify every exam recorded in the table. This kind of identification for exams is also used in other exam-related tables.
examNum	This is used to record of the number of the exam used for this course, so it is easier for admin to organize them. For example, if this is the second exam created for the course “EIE4432”, then examNum is recorded as “2”.

examDate	This is used to record the date in which this exam takes place. This data field is recorded in a string format of (yyyy-mm-dd).
startTime	This is used to record the time in which this exam starts. This data field is recorded in a string of (hour:minute) in 24 hour time format, e.g.“15:30”.
expireTime	This is used to record the time in which this exam starts. This data field is recorded in a string of (hour:minute) in 24 hour time format.
creator	This is used to record the creator of this exam. Only the creator of the exam can grade and publish that exam.
graded	This is used to indicate if the exam is graded and published. “1” indicates it is published while “0” indicates it is not. Only published exams have the statistics and individual papers available for viewing.
remarks	This is used to record any remarks made by the creator regarding this exam.
maxScore	This is used to record the maximum total score achieved by students in this exam. It is set as NULL until the exam is graded.
minScore	This is used to record the minimum total score achieved by students in this exam. It is set as NULL until the exam is graded.
median	This is used to record the median of student total score in this exam. It is set as NULL until the exam is graded.
average	This is used to record the average of student total score in this exam. It is set as NULL until the exam is graded.

exam_questions

`exam_questions` is a table used to store all questions set to all created exams. The following is the statement used to construct this table:

```
$createTable3 = "CREATE TABLE exam_questions
(course varchar(64) NOT NULL,
examNum INT NOT NULL,
QID INT NOT NULL,
QType INT NOT NULL,
score INT NOT NULL,
question varchar(512) NOT NULL,
answer INT,
mc1 varchar(64), mc2 varchar(64), mc3 varchar(64), mc4 varchar(64),
percent DOUBLE(5,2), average DOUBLE(5,2),
PRIMARY KEY (course, examNum, QID))";
```

Data Field	Usage and Description
course	This is used to identify the course of this exam is designed for.
examNum	This is used to record of the number of the exam used for this course.
QID	This is used to identify the question number of this record. “course”, “examNum” and “QID” are used together to uniquely identify records in this table. For example, question 3 of the second exam in EIE4432 will be stored as “course”=“EIE4432”, “examNum”=2,“QID”=3.
QType	This is used to identify the type of this question. “1” = Multiple Choice, “2” = True or False “3” = Short Answer, “4” = Fill in the blanks
score	This is used to record the maximum score achievable in this question. The system will set this score as the score for student answer if the answer is shown correct in automatic grading, or the score is set as the maximum score teacher can give to an answer in manual grading.
question	This is used to record the question string of the question. For fill in the blanks questions, “__BLANK__” is used as an indicator for an input blank, which will be exchanged with an input text field by the system when displaying the question.
answer	This is used to record the answer for multiple choice and true false questions. The system will use this answer to grade the student submissions automatically. The answer is stored as an integer indicating the choice number.(“0” = false, “1” = true) This is left as NULL for short answer and fill in the blanks questions.
mc1	This is used to record the string of the first choice of multiple choice question. This is left as NULL for questions with any question type other than multiple choice, or if the teacher decide to not display this choice.
mc2	This is used to record the string of the second choice of multiple choice question. It is left as NULL for conditions described above.
mc3	This is used to record the string of the third choice of multiple choice question. It is left as NULL for conditions described above.
mc4	This is used to record the string of the fourth choice of multiple choice question. It is left as NULL for conditions described above.
percent	This is used to record the correct percentage of the students’ attempt. It is set as NULL until the exam is graded.
average	This is used to record the average score of the students’ attempt. It

	is set as NULL until the exam is graded.

exam_answers

`exam_answers` is a table which contains all the students' answer to all questions of any exam. The following is the statement used to construct this table:

```
$createTable4 = "CREATE TABLE exam_answers
(course varchar(64) NOT NULL,
examNum INT NOT NULL,
studentID varchar(64) NOT NULL,
QID INT NOT NULL,
answer varchar(512),
score DOUBLE(10,2),
submitTime varchar(64),
PRIMARY KEY (course, examNum, studentID, QID))";
```

Data Field	Usage and Description
course	This is used to identify the course of this exam is designed for.
examNum	This is used to record of the number of the exam used for this course.
studentID	This is used to identify the student who is responsible for this answer.
QID	This is used to identify the question number of this record. "course", "examNum", "studentID" and "QID" are used together to uniquely identify records in this table. For example, a student with user ID "s1" answered the question 3 of the second exam in EIE4432 will be stored as "course"="EIE4432", "examNum"=2, "studentID"="s1","QID"=3.
answer	This is used to record the answer string of this record. If the question type of the question is multiple choice or true false question, this will be stored as a string of integer indicating his choice, e.g."4" = choice 4. If it is a short question, the answer string will be stored instead. For fill in the blanks questions, "__BREAK__" is used as a separator for different answers in different blanks.
score	This is used to record the score rewarded to this answer. If it is graded by the system automatically, the record should either be full mark of the question or zero. If it is waiting to be graded manually, it will be marked as NULL until it is graded.
submitTime	This is used to record the submit time of this answer. Usually it is the same for all answers in the same paper submitted by the same

	student.

Conclusion

To conclude, we have learnt the practical usage of PHP by building such a large project. PHP is so useful when it comes to handling Web cookies and SQL. We have also learnt that Bootstrap is a powerful tool for handling frontend matters. The javascript library JQuery enables us to write javascript functions easier because it has many useful predefined methods that handle complex web tasks. Our overall problem solving skills and experience of developing a web system has been trained by this project. We think this project has a huge potential to be further extended to a business level project, by adding some extra functions and improving the security level of the site. Code refactoring might also be needed when extending this project.